THE DEGREE OF A FINITE LOCALLY FREE MODULE OF CONSTANT RANK ON A PROPER CURVE EQUALS THE DEGREE OF ITS FIRST CHERN CLASS CAPPED WITH THE FUNDAMENTAL CLASS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Degree of an invertible sheaf on a proper curve and its behaviour under components, modification and tensor product. Varieties, labels varieties-definition-degree-invertible-sheaf (statement, proof line 8107), varieties-lemma-degree-in-terms-of-components (proof line 8117), varieties-lemma-degree-birational-pullback (proof line 8131), varieties-lemma-degree-additive (proof line 8135), varieties-lemma-degree-tensor-product (proof line 8150), varieties-lemma-degree-effective-Cartier-divisor (proof line 8155) and varieties-lemma-dim-1-proper-projective (proof line 8148). Supplies the notion of degree being compared, its decomposition over the irreducible components of X, its invariance under a birational modification, its additivity in a filtration and in tensor products, and projectivity of a proper curve; none of them contains the comparison with c_1 itself.
- Filtering a finite locally free sheaf on a curve by invertible quotients after a modification. Divisors, labels divisors-lemma-filter-after-modification (proof line 8122) and divisors-lemma-quasi-projective-Noetherian-pic-effective-Cartier (proof line 8143). Gives the modification X' -> X over which f^*E has a filtration with invertible successive quotients, and expresses an invertible sheaf on a proper curve as O(D) tensor O(D')^{-1} for effective Cartier divisors.
- Compatibility of first Chern classes with pushforward, additivity and the geometric case. Same chapter chow.tex, labels lemma-pushforward-cap-c1 (proof line 8126), lemma-additivity-chern-classes (proof line 8136), lemma-c1-cap-additive and lemma-chern-classes-E-tensor-L (proof line 8152) and

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lemma-geometric-cap (proof line 8157); Algebra, label algebra-lemma-pushdown-module (proof line 8172). Moves the computation across the modification, splits it additively over the filtration, reduces the invertible case to O(D), identifies c_1(O(D)) cap [X] with [D]_0, and converts the resulting sum of residue degrees into the dimension of the space of global sections of O_D.

Source reference. Original lemma 02M0: sources/stacks-chow/tags.txt, source lines 3427–3427. Navigation pointer for \ref{algebra-lemma-pushdown-module} used inside the counted span (line 8172); the argument that consumes it is retained in the span.

Source reference. Original lemma 0AYP: sources/stacks-chow/tags.txt, source lines 13797–13797. Navigation pointer for \ref{divisors-lemma-filter-after-modification} used inside the counted span (line 8122); the argument that consumes it is retained in the span.

Source reference. Original lemma 0AYM: sources/stacks-chow/tags.txt, source lines 13795–13795. Navigation pointer for \ref{divisors-lemma-quasi-projective-Noetherian-pic-effective-Cartier} used inside the counted span (line 8143); the argument that consumes it is retained in the span.

Source reference. Original lemma 02UI: sources/stacks-chow/pinned-original-source.tex, source lines 7965–7965. Navigation pointer for \ref{lemma-additivity-chern-classes} used inside the counted span (line 8136); the argument that consumes it is retained in the span.

Source reference. Original lemma 02SP: sources/stacks-chow/pinned-original-source.tex, source lines 4330–4330. Navigation pointer for \ref{lemma-c1-cap-additive} used inside the counted span (line 8152); the argument that consumes it is retained in the span.

Source reference. Original lemma 02UD: sources/stacks-chow/pinned-original-source.tex, source lines 7755–7755. Navigation pointer for \ref{lemma-chern-classes-E-tensor-L} used inside the counted span (line 8152); the argument that consumes it is retained in the span.

Source reference. Original lemma 02SQ: sources/stacks-chow/pinned-original-source.tex, source lines 4432–4432. Navigation pointer for \ref{lemma-geometric-cap} used inside the counted span (line 8157); the argument that consumes it is retained in the span.

Source reference. Original lemma 02U9: sources/stacks-chow/pinned-original-source.tex, source lines 7394–7394. Navigation pointer for \ref{lemma-pushforward-cap-cj} used inside the counted span (line 8126); the argument that consumes it is retained in the span.

Source reference. Original lemma 0AYR: sources/stacks-chow/tags.txt, source lines 13799–13799. Navigation pointer for \ref{varieties-definition-degree-invertible-sheaf} used inside the counted span (line 8107); the argument that consumes it is retained in the span.

Source reference. Original lemma 0AYS: sources/stacks-chow/tags.txt, source lines 13800–13800. Navigation pointer for \ref{varieties-lemma-degree-additive} used inside the counted span (line 8135); the argument that consumes it is retained in the span.

Source reference. Original lemma 0AYU: sources/stacks-chow/tags.txt, source lines 13802–13802. Navigation pointer for \ref{varieties-lemma-degree-birational-pullback}

used inside the counted span (line 8131); the argument that consumes it is retained in the span.

Source reference. Original lemma 0AYY: sources/stacks-chow/tags.txt, source lines 13806–13806. Navigation pointer for \ref{varieties-lemma-degree-effective-Cartier-divisor} used inside the counted span (line 8155); the argument that consumes it is retained in the span.

Source reference. Original lemma 0AYW: sources/stacks-chow/tags.txt, source lines 13804–13804. Navigation pointer for \ref{varieties-lemma-degree-in-terms-of-components} used inside the counted span (line 8117); the argument that consumes it is retained in the span.

Source reference. Original lemma 0AYX: sources/stacks-chow/tags.txt, source lines 13805–13805. Navigation pointer for \ref{varieties-lemma-degree-tensor-product} used inside the counted span (line 8150); the argument that consumes it is retained in the span.

Source reference. Original lemma 0A26: sources/stacks-chow/tags.txt, source lines 12694–12694. Navigation pointer for \ref{varieties-lemma-dim-1-proper-projective} used inside the counted span (line 8148); the argument that consumes it is retained in the span.

Lemma 0.1. *Let k be a field. Let X be a proper scheme over k of dimension ≤ 1. Let $\mathcal{E}$ be a finite locally free $\mathcal{O}_X$-module of constant rank. Then*

$$\deg(\mathcal{E}) = \deg(c_1(\mathcal{E}) \cap [X]_1)$$

where the left hand side is defined in Varieties, Definition 0AYR.

Proof. Let $C_i \subset X$, $i = 1, \ldots, t$ be the irreducible components of dimension 1 with reduced induced scheme structure and let m_i be the multiplicity of C_i in X. Then $[X]_1 = \sum m_i[C_i]$ and $c_1(\mathcal{E}) \cap [X]_1$ is the sum of the pushforwards of the cycles $m_i c_1(\mathcal{E}|_{C_i}) \cap [C_i]$. Since we have a similar decomposition of the degree of $\mathcal{E}$ by Varieties, Lemma 0AYW it suffices to prove the lemma in case X is a proper curve over k.

Assume X is a proper curve over k. By Divisors, Lemma 0AYP there exists a modification $f : X' \to X$ such that $f^*\mathcal{E}$ has a filtration whose successive quotients are invertible $\mathcal{O}_{X'}$-modules. Since $f_*[X']_1 = [X]_1$ we conclude from Lemma 02U9 that

$$\deg(c_1(\mathcal{E}) \cap [X]_1) = \deg(c_1(f^*\mathcal{E}) \cap [X']_1)$$

Since we have a similar relationship for the degree by Varieties, Lemma 0AYU we reduce to the case where $\mathcal{E}$ has a filtration whose successive quotients are invertible $\mathcal{O}_X$-modules. In this case, we may use additivity of the degree (Varieties, Lemma 0AYS) and of first Chern classes (Lemma 02UI) to reduce to the case discussed in the next paragraph.

Assume X is a proper curve over k and $\mathcal{E}$ is an invertible $\mathcal{O}_X$-module. By Divisors, Lemma 0AYM we see that $\mathcal{E}$ is isomorphic to $\mathcal{O}_X(D) \otimes \mathcal{O}_X(D')^{\otimes -1}$ for some effective Cartier divisors D, D' on X (this also uses that X is projective, see Varieties, Lemma 0A26 for example). By additivity of degree under tensor product of invertible sheaves (Varieties, Lemma 0AYX) and additivity of c_1 under tensor product of invertible sheaves (Lemma 02SP or 02UD) we reduce to the case $\mathcal{E} = \mathcal{O}_X(D)$. In this case the left hand side gives $\deg(D)$ (Varieties, Lemma 0AYY) and the right hand side gives $\deg([D]_0)$ by Lemma 02SQ. Since

$$[D]_0 = \sum\nolimits_{x \in D} \text{length}_{\mathcal{O}_{X,x}}(\mathcal{O}_{D,x})[x] = \sum\nolimits_{x \in D} \text{length}_{\mathcal{O}_{D,x}}(\mathcal{O}_{D,x})[x]$$

by definition, we see

$$\deg([D]_0) = \sum\nolimits_{x \in D} \text{length}_{\mathcal{O}_{D,x}}(\mathcal{O}_{D,x})[\kappa(x) : k] = \dim_k \Gamma(D, \mathcal{O}_D) = \deg(D)$$

The penultimate equality by Algebra, Lemma 02M0 using that D is affine. $\square$